PeopleTec Pi Day Competition

Raspberry Pi Project

Summer Phillips

Hardware

Raspberry Pi 3 Model B

20x4 LCD Display with I2C backpack

3D printed case

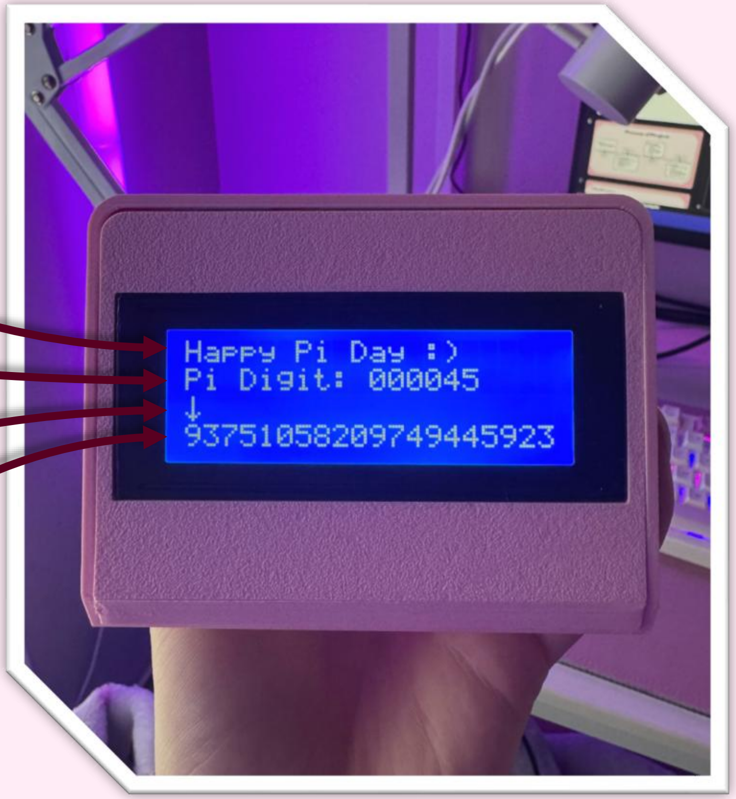
Demonstration

Line 1
Happy Pi Day message

Line 2
Current Pi digit index

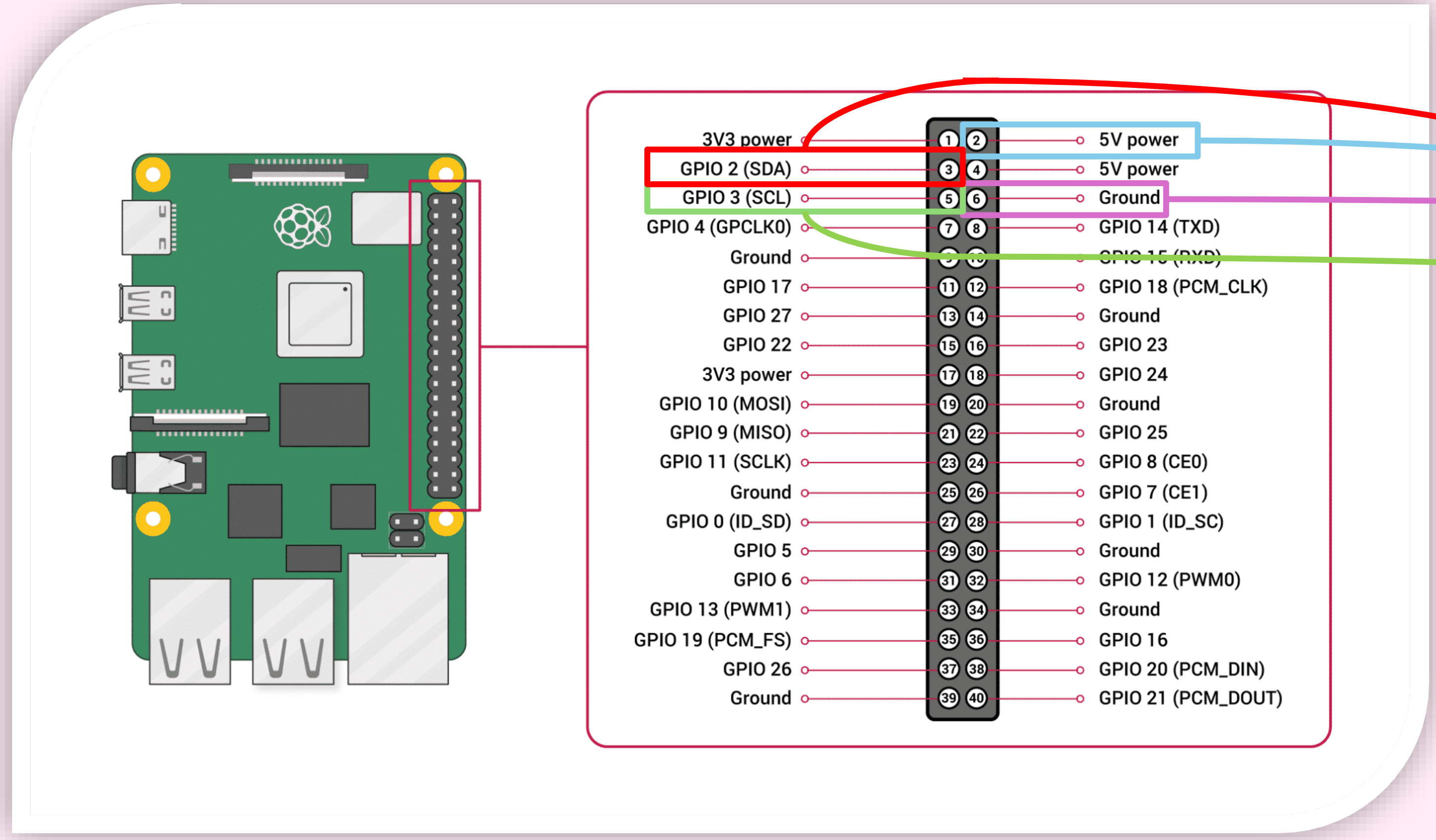
Line 3
Arrow marks the current digit

Line 4
Scrolling digits of π



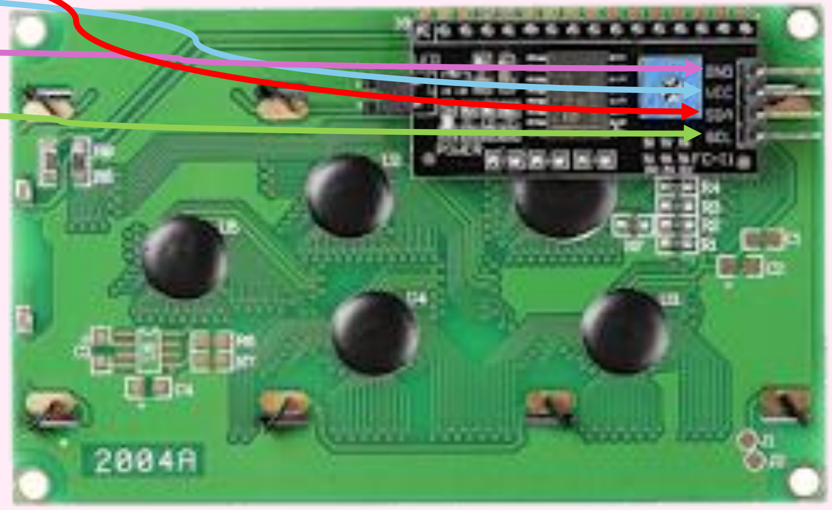
The display refreshes every 100 digits and briefly shows the "Happy Pi Day" message.

Process of Project - Hardware

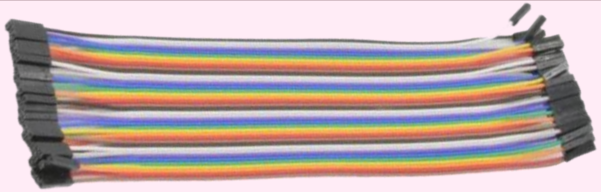


Raspberry Pi pin Layout

LCD Screen



Dupoint Jumper Cables (Female to Female)



Process of Project

Step 1

Hardware
Wired LCD display to Raspberry Pi

Step 2

Communication
Enabled I2C protocol
Installed I2C diagnostic tools
Verified LCD detection

Step 3

Programming
Used ChatGPT to help write Python script to control LCD

Step 4

Automation
Created a systemd service
Program runs automatically when device powers on

Challenges

1.

Problem:
Flickering Screen

Cause:
Issues keeping the jumper wires firmly connected

Solution:
Hot glue the wires to the raspberry pi base

2.

Problem:
Configuring the I2C

Cause:
The I2C interface was not detected

Solution:
Enable the I2C protocol in the settings

The End

☺

This presentation ends here... but Pi keeps going.